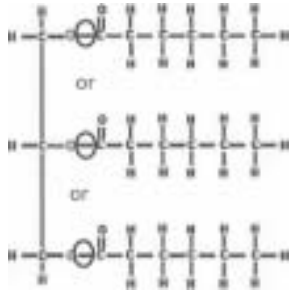


2(a)(i)	circle around any one correct bond ; R if circle includes the double bond O	1
		
2(a)(ii)	condensation ; A dehydration (synthesis) A esterification three ;	2
2(b)(i)	macromolecule <u>and</u> extracellular enzyme ;	1
2(b)(ii)	4 (hours) ;	1
2(b)(iii)	any four from: 1 triglyceride hydrolysis / action of lipase, produces <u>fatty acids</u> ; 2 decrease in pH / described, because of production of (fatty) acids ; 3 steepest rate of (pH) decrease at start because, substrate / triglyceride, concentration highest ; 4 rate of pH decrease / AW, slows because decrease in rate of formation of fatty acids ; 5 rate of pH decrease / AW, slows because enzymes, not at optimum / <u>begin to denature</u> ; 6 rate of pH decrease slows because, substrate / triglyceride, being used up or substrate concentration becomes a limiting factor / AW ; 7 <i>ref. to</i> plateau at 18 hours / at pH 6.1, enzymes are denatured ; 8 <i>ref. to</i> plateau at 18 hours / at pH 6.1, as all the, triglyceride / substrate, has been, hydrolysed / AW ; 9 optimum pH for lipase activity, is pH 8 / alkaline conditions / at the start ; 10 AVP ; e.g. <i>ref. to</i> tertiary structure of lipase	4

Question	Answer	Marks
2(b)(iv)	any two from: (initial) decrease in pH would be more rapid ; plateau would be reached earlier ; AW plateau would be at the same pH (as the previous experiment) ;	2

Question	Answer	Marks
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